NAPAC 2025 · JACOW PROCEEDINGS · SACRAMENTO, CA

AtomicAndPhysicalConstants.jl: A Fast Julia Package to Access Particle Properties and Fundamental Constants

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1 PROBLEM

- **SciBmad**, a new Julia environment for charged-particle beam simulation, needs fast, stable access to physical constants.
- Data are spread across **CODATA, NIST, and PDG**, and simulations need different unit systems and numeric types.
- Converting units at run time adds work and makes functions **type-unstable**, so the compiler cannot optimize them.

2 METHOD

```
using AtomicAndPhysicalConstants
# configure once: units + numeric type
@APCdef unitsystem = ACCELERATOR unittype = Float

APC.C_LIGHT              # m/s
→ 2.99792458e8
e = Species("electron")  # or "H+"
massof(e)                # eV/c²
→ 510998.95069
```

CONFIGURE Choose units and numeric type once → **GENERATE** Macro writes fixed-type constants and getters → **RUN** No conversion; types known at compile time

One `@APCdef` call generates the code; `Species` wraps mass, charge, spin, and moment for particles and isotopes.

3 KEY RESULTS

< 1 µs

Core function run time, including `@APCdef` and `Species()`; all exported functions verified type-stable with JET.jl

16 + 14

Common constants and subatomic particles, plus all NIST isotope data; sourced from CODATA, NIST, and PDG

3 × 3

Unit systems × numeric types (Float64, Unitful, DynamicQuantities), chosen in a single macro call

doi.org/10.18429/JACoW-NAPAC2025-MOP010 · github.com/bmad-sim/AtomicAndPhysicalConstants.jl